No infinite cluster at criticality for long-range percolation with fast-decaying interactions

Yan Ru Pei

yanrpei@gmail.com

October 2026

Abstract

We consider independent long-range bond percolation on $\mathbb{Z}^d$ in which x and y are joined with probability $1 - e^{-\beta J(x-y)}$. Let $d \geq 2$ and let J be invariant under coordinate permutations and sign changes, positive on nearest-neighbour pairs, and satisfy $\liminf_{n\to\infty} n^d \sum_{\|x\|>n} J(x) = 0$. We prove that there is no infinite cluster at the critical parameter. Together with Berger's theorem for heavy tails, this gives continuity of the phase transition for $J(x) \asymp \|x\|^{-d-\alpha}$ for every $\alpha > 0$ except $\alpha = d$, in every dimension $d \geq 2$, and for every exponentially decaying kernel satisfying these hypotheses. The proof adapts a recent renormalization argument for nearest-neighbour percolation on $\mathbb{Z}^3$ that is driven by a gluing inequality conjectured by Kozma and Nitzan. Long edges enter in two places: they can jump over the surfaces that a nearest-neighbour path must cross, and they let a cluster outside a box enter it from far away. We control the first with thick outer faces and the tail condition, and the second by measuring entrances through the total edge weight between a cluster and a box rather than through the number of boundary vertices. The proof is formally verified in Lean 4.

1 Introduction

1.1 Long-range percolation and the main result

Let $d \geq 1$ and let $J : \mathbb{Z}^d \setminus \{0\} \to [0, \infty)$ be a kernel with $J(x) = J(-x)$ and $\sum_x J(x) < \infty$. In *long-range percolation* at parameter $\beta > 0$, each unordered pair $\{x, y\}$ of distinct vertices of $\mathbb{Z}^d$ is open independently with probability

$$p_{xy}(\beta) = 1 - e^{-\beta J(x-y)},$$

and closed otherwise; pairs with $J(x-y) = 0$ are never open. Write $\mathbb{P}_\beta$ for the law, $\theta(\beta) = \mathbb{P}_\beta(0 \leftrightarrow \infty)$ for the probability that the origin lies in an infinite open cluster, and $\beta_c = \inf\{\beta : \theta(\beta) > 0\}$. The function θ is nondecreasing, and the basic question is whether it is continuous at β_c, that is, whether $\theta(\beta_c) = 0$.

For nearest-neighbour percolation, $\theta(p_c) = 0$ was proved in $d = 2$ by Harris and Kesten [11, 14] and in high dimensions by the lace expansion [10, 7]. Kozma and Nitzan [15] reduced the remaining dimensions to a correlation inequality for independent percolation on finite graphs, which has since been proved in several forms [16, 17, 4, 8]. We use it in the joint form of Theorem 2.1, which holds for arbitrary edge probabilities; all graphs to which we apply it are finite.

For long-range models the answer depends on the tail of J. Writing

$$T(n) = \sum_{\|x\|>n} J(x), \qquad \|x\| = \max_i |x_i|,$$

we use the following hypotheses.

(Sym) J is invariant under coordinate permutations and sign changes of coordinates.

(Pos) $J(e_1) > 0$, where e_1 is the first unit vector.

(Tail) $\liminf_{n\to\infty} n^d T(n) = 0$.

Theorem 1.1. *Let $d \geq 2$ and let J satisfy* (Sym), (Pos) *and* (Tail). *Then $0 < \beta_c < \infty$ and $\mathbb{P}_{\beta_c}(0 \leftrightarrow \infty) = 0$.*

Combined with Berger's theorem for heavy-tailed kernels [3, Theorem 2.6], which we recall in Section 7, this settles the power-law case away from the borderline exponent.

Corollary 1.2. *Let $d \geq 2$, $\alpha > 0$, $\alpha \neq d$, and let J satisfy* (Sym) *and $c\|x\|^{-d-\alpha} \leq J(x) \leq C\|x\|^{-d-\alpha}$ for all $x \neq 0$, with $0 < c \leq C < \infty$. Then $\theta(\beta_c) = 0$.*

Corollary 1.3. *Let $d \geq 2$ and let J satisfy* (Sym), (Pos) *and $J(x) \leq Ae^{-a\|x\|}$ for some $A, a > 0$. Then $\theta(\beta_c) = 0$. The same holds for every finite-range J satisfying* (Sym) *and* (Pos).

Corollary 1.2 covers both regimes of α around the exceptional value d. For $\alpha < d$ it is Berger's theorem; for $\alpha > d$ it follows from Theorem 1.1, since then $T(n) \leq C'n^{-\alpha}$. The value $\alpha = d$ is excluded: there $n^d T(n)$ is bounded away from zero, so edges of length comparable to n leave the cube $[-n, n]^d$ with probability of order one at every scale. This is the mechanism behind the discontinuous transition of the one-dimensional model with $J(x) \asymp |x|^{-2}$ [1, 5], and Theorem 1.1 says nothing about it.

1.2 Earlier results

Berger [3, Theorem 2.6] proved that $\theta(\beta_c) = 0$ whenever $\liminf_{\|k\|\to\infty} \mathbb{P}(0 \leftrightarrow k \text{ directly})\, \|k\|_1^s > 0$ for some $s < 2d$, in every $d \geq 1$; Hutchcroft [13] made this quantitative. In the opposite regime the kernel decays fast enough that the model should behave like nearest-neighbour percolation, and Hutchcroft writes that "it is widely believed that the phase transition should be continuous for all $\alpha > 0$ and $d \geq 2$" [13, p. 2]. The known cases of this regime are the following.

- In $d = 2$, Duminil-Copin, Sidoravicius and Tassion [6] proved continuity for finite-range kernels invariant under the symmetries of $\mathbb{Z}^2$, and Angel, de la Riva, Panagiotis and Shi [2] extended this to kernels with exponential decay; they state the case $\alpha > 2$ of power laws in $d = 2$ as a hope [2, §1.4]. Both arguments are planar in spirit: they glue clusters that cross as subsets of $\mathbb{R}^2$.
- In high dimensions, the lace expansion gives mean-field behaviour, and in particular $\theta(\beta_c) = 0$, for sufficiently spread-out kernels when $d > 3(\alpha \wedge 2)$ [12].
- For kernels that are uniform on a finite range, Theorem 1.1's conclusion follows from the recent preprint on quasi-transitive graphs [18].

Theorem 1.1 treats every $d \geq 2$ and requires neither planarity nor high dimension nor a spread-out kernel. It contains the theorem of [2], whose kernels satisfy (Sym) (in $d = 2$, invariance under $\pi/2$-rotations and reflections), (Pos) and exponential decay.

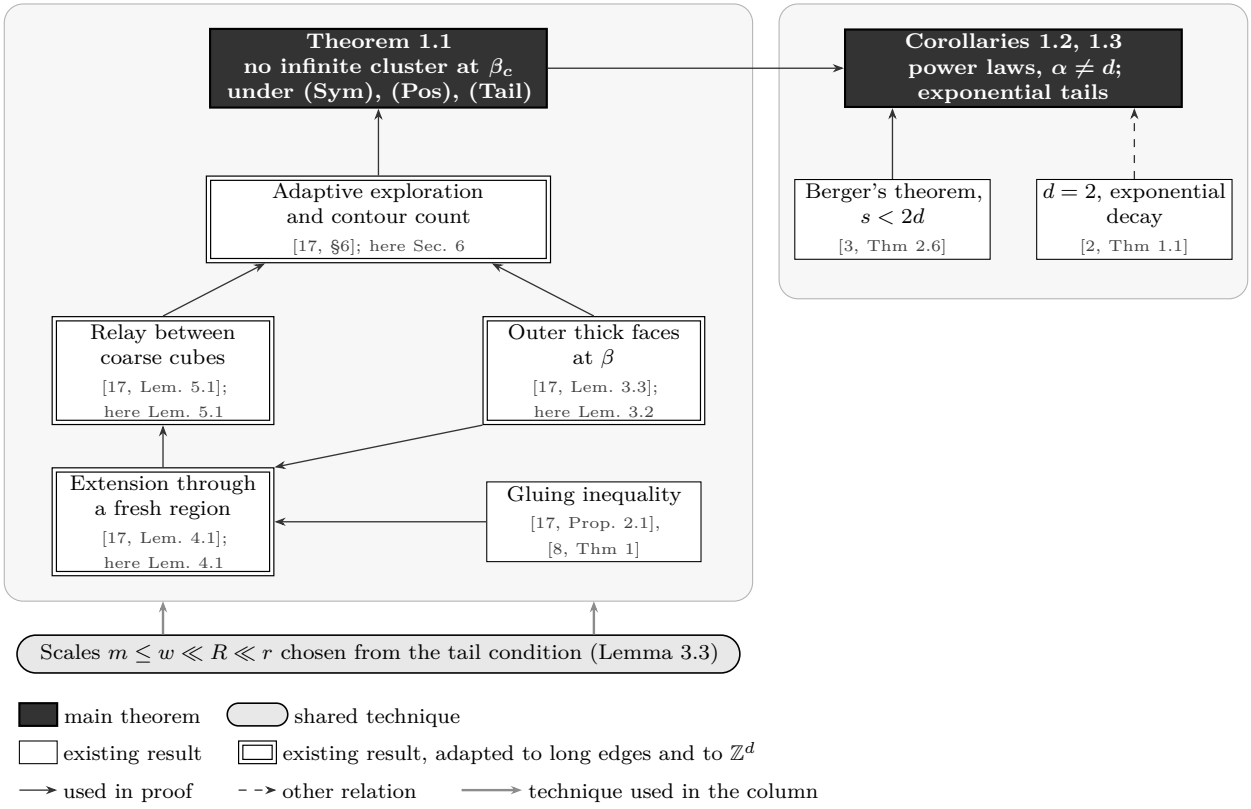


Figure 1: How Theorem 1.1 and its corollaries are proved. A solid arrow means that its source is used in the proof of its target; only the main dependencies are drawn. The dashed arrow means that the theorem of [2] is contained in the case $d = 2$ of Corollary 1.3. Box styles give provenance; grey labels give the source of each adapted step and where it is proved here. The modifications are listed in Section 1.3.

1.3 Idea of the proof

We follow the self-contained proof of $\theta(p_c) = 0$ for nearest-neighbour percolation on $\mathbb{Z}^3$ in [17], which builds on the renormalization scheme of Kozma and Nitzan [15, §4] and the seed-and-exploration method of Grimmett and Marstrand [9]. Figure 1 shows the structure. Assume, for a contradiction, that $\theta(\beta) > 0$ at $\beta = \beta_c$. The argument has four steps.

1. *Faces at β.* A large box $\Lambda_m(u)$ connects with high probability, inside a surrounding cube $\Lambda_\ell(u)$, to each of the $d2^d$ quarter-faces of that cube (a face cut in half in every transverse coordinate). Only three radii ℓ are needed, so these finitely many estimates persist at some $q < \beta$.

2. *Extension.* If a vertex o outside a box B reaches B, then it reaches, with almost the same probability, any target set T that every point near B can hit through a quarter-face. The cluster of o enters a neighbourhood of B at many possible points; at each one a small *seed* box, if all its edges are open, gives a *relay point* that reaches T with high probability. The gluing inequality (Theorem 2.1) turns "every relay point reaches T with high probability" into "o reaches T if it reaches a relay point".

3. *Relay.* A fixed sequence of extensions moves a connection from the centre of one coarse cube to the centre of a neighbouring one.

4. *Exploration.* Cubes indexed by $\mathbb{Z}^2$ are explored one at a time, keeping a high conditional connection probability for each pending cube; a planar contour count then yields an infinite cluster at $q < \beta_c$, a contradiction.

Long edges break two nearest-neighbour features of this argument.

First, a path that leaves a box need not pass through its boundary: its first step outside can be a long jump. In Step 1 we therefore use *outer thick quarter-faces*, slabs of width w just outside a face of the cube (Figure 2), and we require the last edge of the connection to have length at most w. Jumps longer than w out of Λ_ℓ have total probability at most $\beta(2\ell+1)^d T(w)$. The tail condition makes this small for a w that is a small fixed fraction of ℓ, so the thick faces still fit inside the relay geometry of Step 3.

Second, in Step 2 the nearest-neighbour proof selects an expansion $B' = B^{(j)}$ of B such that the cluster of o outside B' has many exterior neighbours, using the fact that closing the single edge from each exterior neighbour into B' disconnects o from B'. With long edges, every vertex of the outside cluster has edges into B'. We replace the count of exterior neighbours by the *crossing weight*

$$W = \sum_{x \in \mathcal{C}} \sum_{z \in B'} \beta J(x - z),$$

where $\mathcal{C}$ is the cluster of o in the complement of B'; the probability that all crossing edges are closed is exactly e^{-W} (Figure 3). If W is small this disconnection is cheap, and disjointness of the corresponding events over well-separated layers shows that a small weight is rare at some layer. If W is large, the landing points of the crossing edges support many independent attempts to open a seed box, because a single seed box $\Lambda_m(v)$ can absorb at most $\beta(2m+1)^d \|J\|_1$ of the weight. Entrances that jump over a shell of width w around B' are again controlled by the tail condition. The other steps of [17] need only bookkeeping: d contraction steps instead of three in the relay between neighbouring cubes, margins enlarged by w, and an exploration whose root cube is opened without fixing the edges that leave it.

The scales are chosen in the order $m \le w \ll R \ll r$, where R and r are large constant multiples of w and the tail condition makes $\beta r^d T(w)$ small (Lemma 3.3). Only one such choice is needed; this is why a lim inf suffices.

Adapted results. Lemma 3.2 and Lemma 3.3 adapt [17, Lemma 3.3] (outer thick faces and short exits); Lemma 4.1 adapts [17, Lemma 4.1], itself adapted from [15, Lemma 10] (crossing weight, layers spaced by w, seeds at landing points); Lemma 5.1 adapts [17, Lemma 5.1] to $\mathbb{Z}^d$ and to the margins $R + w$; Section 6 adapts [17, §6, Lemmas 6.1–6.3] (long-range edges between neighbouring cubes, a hypothesis on edges entering the union of two cubes, and the root). The planar contour count is used as in [17, Lemma 3.1], and we include its proof.

1.4 Formal verification

Theorem 1.1 is formalized in Lean 4 with Mathlib [19]. The formal statement is `Kernel.main`, in the namespace `Peapods.LongRange`. It defines the model, θ and β_c as above, with the hypotheses (Sym), (Pos) and $d \ge 2$. (Tail) is encoded as "$n^d T(n) < \varepsilon$ for infinitely many n, for every $\varepsilon > 0$". This is equivalent to (Tail) because $n^d T(n) \ge 0$, and the formalization also proves it equivalent to the liminf taken in $[0, \infty]$. The parts of the Lean proof of [17] that do not depend on the lattice are used unchanged: product-law conditioning, Harris's inequality, the hyperedge forms of Theorem 2.1 and Corollary 2.2, adaptive reveals, and the planar contour count of Lemma 2.3. The long-range steps (Lemmas 2.4–5.1 and Section 6) are formalized anew, following the structure of that proof. The formal proof uses no unformalized result and depends only on Lean's three standard axioms (propositional extensionality, choice and quotient soundness). Corollaries 1.2 and 1.3 are not formalized.

1.5 Organization

Section 2 collects the inputs. Sections 3–6 carry out Steps 1–4, and Section 7 proves the corollaries and records where each hypothesis is used. Section 8 closes with open problems.

2 Preliminaries

Notation. $\Lambda_b(u) = u + ([-b, b]^d \cap \mathbb{Z}^d)$ and $\Lambda_b = \Lambda_b(0)$. A *rectangle* is a nonempty product of integer intervals, and $B^{(j)}$ denotes the expansion of a rectangle B by j in every coordinate direction. Throughout, $\|J\|_1 = \sum_x J(x)$; it is finite under (Tail), since $T(n_0) < \infty$ for some n_0 and the remaining sum is finite.

Product laws on finite graphs. In Sections 4–6 we work on finite graphs $G = (V, E)$ with $V \subset \mathbb{Z}^d$, whose edges are pairs $\{x, y\}$ with $J(x - y) > 0$, with independent bits of individual probabilities $p_e \in [0, 1]$. We put $\lambda_e = -\log(1 - p_e) \in [0, \infty]$, so that $\mathbb{P}(e \text{ open}) \le \lambda_e$ and the probability that all edges of a set S are closed is $\exp(-\sum_{e \in S} \lambda_e)$. A bit is *internal* to a vertex set U if both endpoints lie in U. For $q > 0$ we say that $(F_q(U))$ holds if every pair $\{x, y\} \subset U$ with $J(x - y) > 0$ is an edge of G with $\lambda_{xy} = qJ(x - y)$.

The gluing inequality.

Theorem 2.1 ([17, 4, 8])**.** *Consider independent bond percolation on a finite graph with arbitrary edge probabilities in* $[0, 1]$*. For vertices* s, t *and a nonempty vertex set* A*,*

$$\mathbb{P}(s \leftrightarrow A,\ s \leftrightarrow t) \ \ge\ \mathbb{P}(s \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow t).$$

This is a joint form of Conjecture 1 of Kozma and Nitzan [15], in which the event $\{s \leftrightarrow t\}$ alone appears on the left. Theorem 2.1 is proved for hypergraphs, with t replaced by a vertex set, in [17, Proposition 2.1], and in the form above in [8, Theorem 1], where the extension from $(0,1)$ to $[0,1]$ is by continuity; it also follows from [4, Theorem 1] together with Harris's inequality. The Lean formalization of Leder [16] proves the additive form: if $\mathbb{P}(a \leftrightarrow t) \geq 1-b$ for every $a \in A$, then $\mathbb{P}(s \leftrightarrow t) \geq \mathbb{P}(s \leftrightarrow A) - b$ [16, Theorem 2.2]. Either form gives the following corollary, which is [17, Corollary 2.3]; we include the short proof from Theorem 2.1.

Corollary 2.2 (failure comparison)**.** *Consider independent bond percolation on a finite graph G with arbitrary edge probabilities in $[0,1]$, let A, T be vertex sets and $b \in [0,1]$. If $\mathbb{P}(a \leftrightarrow T) \geq 1-b$ for every $a \in A$, then $\mathbb{P}(o \leftrightarrow A,\ o \not\leftrightarrow T) \leq b$ for every vertex o. The same holds after any deterministic set of bits has been fixed.*

Proof. If $o \in T$, $A = \varnothing$ or $T = \varnothing$ (which forces $b = 1$ when $A \neq \varnothing$) there is nothing to prove. Otherwise let G' be obtained from G by identifying the vertices of T to a single vertex t, removing loops, and merging the parallel edges $e_1, \dots, e_k$ between two vertices into one edge, open if and only if one of the e_i is open; this couples the two models, and the merged edge has probability $1 - \prod_i (1-p_{e_i})$. Under the coupling, $o \not\leftrightarrow T$ in G if and only if $o \not\leftrightarrow t$ in G', and then the open clusters of o in G and G' coincide. Let A' be the image of A in G'. Hence $\mathbb{P}_G(o \leftrightarrow A,\ o \not\leftrightarrow T) = \mathbb{P}_{G'}(o \leftrightarrow A',\ o \not\leftrightarrow t)$. In G' every $a \in A'$ satisfies $\mathbb{P}_{G'}(a \leftrightarrow t) \geq 1-b$: for $a \notin T$ this is $\mathbb{P}_G(a \leftrightarrow T) \geq 1-b$, and $t \leftrightarrow t$ always. Theorem 2.1 in G' gives $\mathbb{P}_{G'}(o \leftrightarrow A',\ o \not\leftrightarrow t) = \mathbb{P}_{G'}(o \leftrightarrow A') - \mathbb{P}_{G'}(o \leftrightarrow A',\ o \leftrightarrow t) \leq b\,\mathbb{P}_{G'}(o \leftrightarrow A') \leq b$. Fixing a bit means giving it probability 0 or 1. □

We also use Harris's inequality [11]: increasing events of a finite product of independent bits are positively correlated, and so are decreasing events.

A planar boundary count. The following lemma is [17, Lemma 3.1]; we include the short proof.

Lemma 2.3. *There are families $\mathcal{C}_n$ of simple cycles of length n in the dual lattice $\mathbb{Z}^2 + (\frac{1}{2}, \frac{1}{2})$, with $|\mathcal{C}_n| \leq 2n3^n$, such that the dual edge boundary of every finite $X \subset \mathbb{Z}^2$ containing 0 contains a cycle in $\bigcup_n \mathcal{C}_n$. For each cycle in $\mathcal{C}_n$ one can fix at least $n/7$ of its crossed primal edges with pairwise disjoint endpoints.*

Proof. Each dual vertex has even degree in the dual boundary of X (it counts membership changes around a primal square), so the boundary decomposes into edge-disjoint simple cycles. Along the positive horizontal ray from 0 the number of boundary crossings is odd, so some cycle of the decomposition crosses it an odd number of times; a closed cycle crosses the whole horizontal line an even number of times, so this cycle also crosses the negative ray. If its length is n, some positive crossing has horizontal coordinate $t + \frac{1}{2}$ with $0 \leq t < n$. Let $\mathcal{C}_n$ be the set of simple cycles of length n with such an anchored crossing and an odd number of positive crossings: there are at most n anchors, two orientations and 3^n non-backtracking continuations. Finally select crossed primal edges greedily, discarding those incident to an endpoint of a selected edge; each selection discards at most seven edges, since primal degrees are four. □

Elementary facts.

Lemma 2.4. *Under* (Sym), (Pos) *and $\|J\|_1 < \infty$, if $d \geq 2$ then $0 < \beta_c < \infty$. If $\theta(\beta) > 0$, then an infinite cluster exists almost surely, and*

$$\mathbb{P}_\beta(\Lambda_m \text{ meets an infinite cluster}) \to 1 \qquad (m \to \infty). \tag{2.1}$$

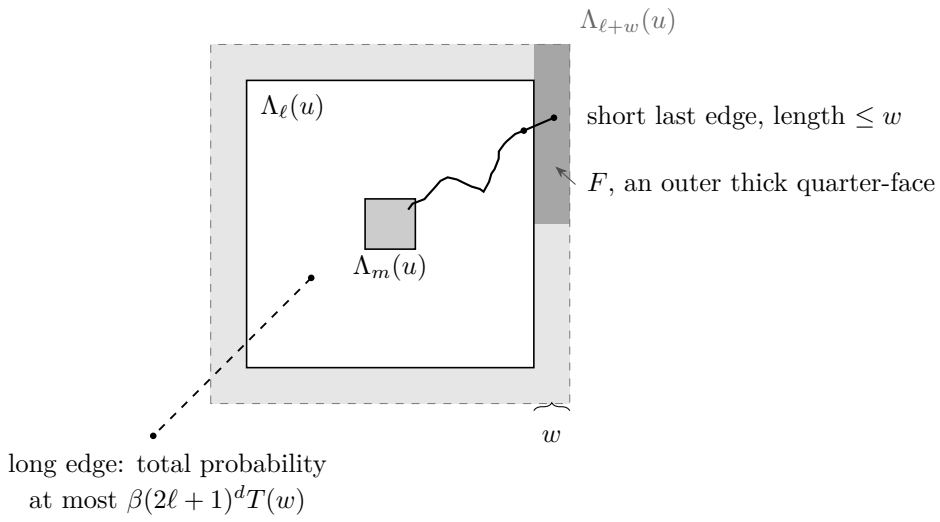


Figure 2: Lemma 3.2 in a planar cross-section. An open path from $\Lambda_m(u)$ that leaves $\Lambda_\ell(u)$ by an edge of length at most w lands in the shell $\Lambda_{\ell+w}(u) \setminus \Lambda_\ell(u)$ (light grey), which the N_F outer thick quarter-faces cover; one is shown in dark grey. A path can also leave by a long edge (dashed), which a nearest-neighbour argument would not see; such edges are rare when $\ell^d T(w)$ is small.

Proof. The nearest-neighbour pairs in a coordinate plane are open independently with probability $1 - e^{-\beta J(e_1)}$, by (Sym) and (Pos), and planar bond percolation with parameter close to 1 percolates by Lemma 2.3 and a union bound over dual cycles; so $\beta_c < \infty$. The cluster of 0 is dominated by a Galton–Watson tree with mean offspring $\sum_y (1 - e^{-\beta J(y)}) \le \beta \|J\|_1$, so $\beta_c \ge 1/\|J\|_1$. Removing an edge splits a connected component into at most two components, so the event that an infinite cluster exists is unchanged by modifying finitely many bits and is a tail event; by Kolmogorov's 0–1 law it has probability 1 when $\theta(\beta) > 0$. The events that Λ_m meets an infinite cluster increase to this event. □

3 Faces at a percolating parameter, and the choice of scales

Definition 3.1. Let $u \in \mathbb{Z}^d$ and $\ell, w \ge 1$. For $i \in \{1, \dots, d\}$, $\sigma \in \{\pm 1\}$ and $\varepsilon = (\varepsilon_j)_{j \ne i} \in \{\pm 1\}^{d-1}$, the *outer thick quarter-face* $F_{i,\sigma,\varepsilon}(u, \ell)$ is

$$\{u + x : \ \sigma x_i \in (\ell, \ell + w], \ 0 \le \varepsilon_j x_j \le \ell + w \ (j \ne i)\}.$$

There are $N_F = d\, 2^d$ of them. For $m < \ell$ and such a face F, let $X(u, \ell, F)$ be the event that there is an open path $x_0, \dots, x_k$ with $x_0 \in \Lambda_m(u)$, $x_0, \dots, x_{k-1} \in \Lambda_\ell(u)$, $x_k \in F$ and $\|x_k - x_{k-1}\| \le w$.

The event $X(u, \ell, F)$ is increasing. It depends only on the bits internal to $\Lambda_\ell(u)$ and on the pairs between $\Lambda_\ell(u)$ and F, all of which are internal to $\Lambda_{\ell+w}(u)$.

Lemma 3.2 (faces at a percolating parameter)**.** *Assume* (Sym) *and* $\theta(\beta) > 0$. *For* $\ell > m$, $w \ge 1$, *every* u *and every outer thick quarter-face* F *of* (u, ℓ),

$$\mathbb{P}_\beta(X(u, \ell, F)^c) \le \Big[\mathbb{P}_\beta(\Lambda_m \textit{ meets no infinite cluster}) + \beta(2\ell+1)^d T(w)\Big]^{1/N_F}.$$

Proof. Take $u = 0$. Let E_{long} be the event that some open edge $\{x, y\}$ has $x \in \Lambda_\ell$ and $\|x - y\| > w$; then $\mathbb{P}(E_{\text{long}}) \le \sum_{x\in\Lambda_\ell} \sum_{\|z\|>w} (1 - e^{-\beta J(z)}) \le \beta(2\ell+1)^d T(w)$. If Λ_m meets an infinite cluster, take a finite open path from Λ_m to a vertex outside Λ_ℓ, and let y be its first vertex outside Λ_ℓ and x its predecessor. Off E_{long} we have $\|x-y\| \le w$, so $y \in \Lambda_{\ell+w} \setminus \Lambda_\ell$. Choose i with $|y_i| > \ell$, $\sigma = \operatorname{sign} y_i$ and $\varepsilon_j = \operatorname{sign} y_j$ (either sign if $y_j = 0$); then $y \in F_{i,\sigma,\varepsilon}(0,\ell)$ and the path up to y witnesses $X(0,\ell,F_{i,\sigma,\varepsilon})$. Hence $\mathbb{P}(\bigcap_F X(0,\ell,F)^c) \le \mathbb{P}(\Lambda_m \text{ meets no infinite cluster}) + \mathbb{P}(E_{\text{long}})$. The events $X(0,\ell,F)^c$ are decreasing and depend on finitely many bits. The signed coordinate permutations fixing 0 act transitively on the faces and, by (Sym), preserve the law, Λ_m and Λ_ℓ; so all N_F complements have the same probability a. Harris's inequality gives $a^{N_F} \le \mathbb{P}(\bigcap_F X(0,\ell,F)^c)$. □

Lemma 3.3 (choice of scales)**.** *Assume* (Sym), (Tail) *and* $\theta(\beta) > 0$. *There are* $f, \delta, \eta, \tau, \gamma \in (0,1)$, *positive integers* m, K, w, R, r, *constants* $\rho_0, A, c_A, L > 0$ *and* $q \in (\beta/2, \beta)$ *such that:*

(S1) $f < \frac{1}{2}$ *and* $\sum_{n\ge1} 2n3^n(2f)^{n/7} < 1$;

(S2) $\delta + 4(d+10)\eta/\delta < f$, $\tau < \eta/4$, $\gamma < \delta$ *and* $2\gamma/\tau < \eta/4$;

(S3) $\mathbb{P}_\beta(\Lambda_m$ *meets no infinite cluster*$) < \frac{1}{2}(\gamma/2)^{N_F}$;

(S4) $\rho_0 = \prod_e (1 - e^{-(\beta/2)J(e)})$, *the product over the pairs* e *internal to* Λ_m *with* $J(e) > 0$; $A = \beta\|J\|_1$; $c_A = (1 - e^{-A})/A$; $4/(K\rho_0) < \eta/4$; *and* $L = K(4m+3)^d/c_A$;

(S5) $\beta(41r+1)^d T(w) \le \min\{\frac{1}{2}(\gamma/2)^{N_F}, \eta/8\}$;

(S6) $w \ge m$, $R \ge m + w + 8e^L w/\eta$, $r > 4(d+12)(R+w+1)$ *and* $r > m + w$;

(S7) *for* $\ell \in \mathcal{L} := \{2r, 2r+1, 10r-w\}$, *every* u *and every outer thick quarter-face* F *of* (u,ℓ), $\mathbb{P}_q(X(u,\ell,F)) > 1 - \gamma$.

Proof. Choose f, then $\delta, \eta, \tau, \gamma$ in this order; then m by (2.1); then ρ_0, A, c_A, K, L. None of these depends on w or r. By (Tail) there are arbitrarily large $w \ge m$ with $w^d T(w)$ as small as we like. For such w put $R = m + w + \lceil 8e^L w/\eta \rceil$ and $r = 4(d+12)(R+w+2)$; then (S6) holds and $r \le C_0 w$ with C_0 depending only on d, m, L, η, so $\beta(41r+1)^d T(w) \le \beta(42C_0)^d w^d T(w)$ and (S5) holds for a suitable such w. Every $\ell \in \mathcal{L}$ satisfies $\ell > m$ and $2\ell + 1 \le 41r + 1$, so Lemma 3.2, (S3) and (S5) give $\mathbb{P}_\beta(X(u,\ell,F)^c) < \gamma/2$. These finitely many events (up to translation) depend on finitely many bits whose probabilities are continuous in the parameter, so (S7) holds for some $q \in (\beta/2, \beta)$. □

We fix these choices from now on, and write $\rho = \prod_e (1 - e^{-qJ(e)}) \ge \rho_0$ (over the same pairs) for the probability that all pairs e internal to a translate of Λ_m with $J(e) > 0$ are open.

4 Extension through a fresh region

Lemma 4.1 (extension)**.** *Let* $G = (V, E)$ *be a finite graph as in Section 2,* $D \subset V$ *a rectangle with* $|D| \le (41r+1)^d$ *and* $(F_q(D))$, B *a rectangle with* $B^{(R)} \subset D$, $o \in V \setminus B^{(R)}$ *and* $T \subset V$. *Assume:*

(E2) *every edge* $\{x, z\}$ *of* G *with* $z \in B^{(R)}$ *and* $x \notin D$ *has* $\lambda_{xz} \le qJ(x - z)$;

(E3) *for every* $u \in B^{(R)}$ *there are* $\ell \in \mathcal{L}$ *and an outer thick quarter-face* F *of* (u,ℓ) *with* $\Lambda_{\ell+w}(u) \subset D$ *and* $F \subset T$.

Then $\mathbb{P}(o \leftrightarrow B,\ o \not\leftrightarrow T) < \eta$.

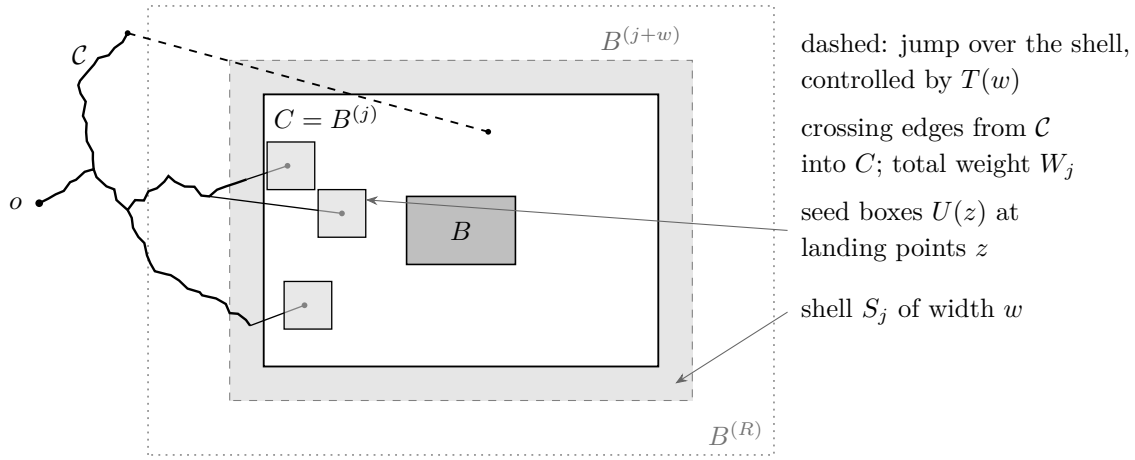


Figure 3: The extension lemma (Lemma 4.1) in a planar cross-section. The cluster $\mathcal{C}$ of o outside $C = B^{(j)}$ reaches the shell S_j. All crossing edges from $\mathcal{C}$ into C are closed with probability exactly e^{-W_j}, which is cheap when W_j is small; when W_j is large, the landing points z in one residue class carry disjoint seed boxes $U(z)$, which give many independent chances to reach a good relay centre. Entrances that jump over the shell are rare by the tail condition.

Proof. By $(F_q(D))$ and (E2), every edge $\{x, z\}$ with $z \in B^{(R)}$ and $x \notin B^{(R)}$ has $\lambda_{xz} \le qJ(x - z)$; hence $\sum_x \lambda_{xz} \le q\|J\|_1 \le A$ for $z \in B^{(R)}$.

Step 1: a layer. For an integer $j \in [m, R)$ let ζ_j be the bits internal to $V \setminus B^{(j)}$, let $\mathcal{C}_j$ be the open cluster of o in the graph on $V \setminus B^{(j)}$ with these bits, and set

$$W_j = \sum_{x \in \mathcal{C}_j} \sum_{z \in B^{(j)}} \lambda_{xz}, \qquad S_j = B^{(j+w)} \setminus B^{(j)},$$

$$L_j = \{W_j < L,\ \mathcal{C}_j \cap S_j \neq \varnothing\}, \qquad \mathrm{Cl}_j = \{\text{every edge between } \mathcal{C}_j \text{ and } B^{(j)} \text{ is closed}\}.$$

Here $W_j \le |B^{(j)}|A < \infty$. The crossing bits between $V \setminus B^{(j)}$ and $B^{(j)}$ are independent of ζ_j, so $\mathbb{P}(\mathrm{Cl}_j \mid \zeta_j) = e^{-W_j} \ge e^{-L}$ on L_j. On Cl_j the open cluster $C(o)$ of o in G equals $\mathcal{C}_j$, because every other edge leaving $\mathcal{C}_j$ is internal to $V \setminus B^{(j)}$ and therefore closed by maximality.

Let $\mathcal{J} = \{m + kw : k \ge 0\} \cap [m, R)$. If $j' < j$ both lie in $\mathcal{J}$, then $j' + w \le j$; on $L_j \cap \mathrm{Cl}_j$ we have $C(o) = \mathcal{C}_j \subset V \setminus B^{(j)}$, hence $\mathcal{C}_{j'} = C(o)$, while $S_{j'} \subset B^{(j)}$, so $\mathcal{C}_{j'} \cap S_{j'} = \varnothing$ and $L_{j'}$ fails. The events $L_j \cap \mathrm{Cl}_j$, $j \in \mathcal{J}$, are therefore pairwise disjoint, and

$$e^{-L} \sum_{j \in \mathcal{J}} \mathbb{P}(L_j) \le \sum_{j \in \mathcal{J}} \mathbb{P}(L_j \cap \mathrm{Cl}_j) \le 1.$$

Since $|\mathcal{J}| \ge (R - m)/w$, we can fix $j \in \mathcal{J}$ with $\mathbb{P}(L_j) \le e^L w/(R - m) \le \eta/8$, by (S6).

Let E_{long} be the event that some open edge $\{x, z\}$ has $z \in B^{(R)}$ and $\|x - z\| > w$. Then $\mathbb{P}(E_{\mathrm{long}}) \le \sum_{z \in B^{(R)}} \sum_{\|x-z\|>w} \lambda_{xz} \le q|D|T(w) \le \eta/8$ by (S5). If $o \leftrightarrow B$, $W_j < L$ and $\mathcal{C}_j \cap S_j = \varnothing$, then an open path from o enters $B^{(j)}$ for the first time at some z from a predecessor $x \in \mathcal{C}_j \setminus B^{(j+w)}$, so $\|x - z\| > w$ and E_{long} occurs. Hence

$$\mathbb{P}(o \leftrightarrow B,\ W_j < L) \le \mathbb{P}(L_j) + \mathbb{P}(E_{\mathrm{long}}) \le \eta/4. \tag{4.1}$$

Step 2: relay quality. Put $C = B^{(j)}$ and let ξ be the bits internal to C. Every side of C has at least $2j + 1 \ge 2m + 1$ points. For $z \in C$ let $v(z)$ be obtained from z by clamping each coordinate to

the corresponding side of C shrunk by m at both ends, let $U(z) = \Lambda_m(v(z)) \subset C$ (so $z \in U(z)$), let $O(z)$ be the event that all bits internal to $U(z)$ are open, so $\mathbb{P}(O(z)) = \rho$, and put

$$g(z) = \mathbb{P}(v(z) \leftrightarrow T \mid \xi), \qquad A_\xi = \{v(z) : z \in C,\ g(z) > 1 - \tau\}.$$

A_ξ is a function of ξ. Fix z. By (E3) at $u = v(z) \in B^{(R)}$ there are $\ell \in \mathcal{L}$ and F with $\Lambda_{\ell+w}(v(z)) \subset D$ and $F \subset T$; the event $X(v(z), \ell, F)$ uses only bits internal to D, so by $(F_q(D))$ and (S7) its probability exceeds $1 - \gamma$. By (Pos) and (Sym) an open box is connected through its nearest-neighbour pairs, so on $O(z) \cap X(v(z), \ell, F)$ we have $v(z) \leftrightarrow T$. Harris's inequality gives $\mathbb{P}(O(z),\ v(z) \not\leftrightarrow T) \le \rho\gamma$, and since $O(z)$ is ξ-measurable,

$$\tau\, \mathbb{P}(O(z),\ g(z) \le 1 - \tau) \le \mathbb{E}[\mathbf{1}_{O(z)}(1 - g(z))] = \mathbb{P}(O(z),\ v(z) \not\leftrightarrow T) \le \rho\gamma. \tag{4.2}$$

Step 3: entrances. Condition on $\zeta = \zeta_j$; write $\mathcal{C} = \mathcal{C}_j$ and $a_z = \sum_{x \in \mathcal{C}} \lambda_{xz}$ for $z \in C$, so that $W_j = \sum_{z \in C} a_z$ and $a_z \le A$. Among the $(4m+3)^d$ residue classes of C modulo $4m+3$, choose (as a function of ζ) a class Z with $\sum_{z \in Z} a_z \ge W_j/(4m+3)^d$. Distinct $z, z' \in Z$ satisfy $\|z - z'\| \ge 4m+3$, and clamping moves each coordinate by at most m, so $\|v(z) - v(z')\| \ge 2m+3$ and the boxes $U(z)$, $z \in Z$, are disjoint.

Let Σ_z be the event that some edge between $\mathcal{C}$ and z is open and $O(z)$ holds. Given ζ, the crossing bits are independent of ξ, edges into distinct z are distinct, and the events $O(z)$, $z \in Z$, use disjoint bits of ξ; hence the events Σ_z, $z \in Z$, are independent with $\mathbb{P}(\Sigma_z \mid \zeta) = \rho(1 - e^{-a_z})$. Let $S = \sum_{z \in Z} \mathbf{1}_{\Sigma_z}$ and $\mu = \sum_{z \in Z}(1 - e^{-a_z})$. Since $a \mapsto (1 - e^{-a})/a$ is decreasing, $1 - e^{-a_z} \ge c_A a_z$ and $\mu \ge c_A W_j/(4m+3)^d$, which is at least K on $\{W_j \ge L\}$. By Chebyshev's inequality, on $\{W_j \ge L\}$,

$$\mathbb{P}(S < \rho\mu/2 \mid \zeta) \le \frac{4}{\rho\mu} \le \frac{4}{\rho_0 K} < \frac{\eta}{4}.$$

Let $Z_{\mathrm{bad}} = \sum_{z \in Z} \mathbf{1}_{\Sigma_z} \mathbf{1}\{g(z) \le 1 - \tau\}$. By the independence of the crossing bits from ξ and (4.2), $\mathbb{E}[Z_{\mathrm{bad}} \mid \zeta] = \sum_z (1 - e^{-a_z})\mathbb{P}(O(z), g(z) \le 1 - \tau) \le \mu\rho\gamma/\tau$, so $\mathbb{P}(Z_{\mathrm{bad}} \ge \rho\mu/2 \mid \zeta) \le 2\gamma/\tau < \eta/4$. If $S \ge \rho\mu/2 > Z_{\mathrm{bad}}$, some $z \in Z$ has Σ_z and $g(z) > 1 - \tau$: then o reaches some $x \in \mathcal{C}$, the open edge $\{x, z\}$, and $v(z) \in A_\xi$ through the open box $U(z)$. Together with (4.1),

$$\mathbb{P}(o \leftrightarrow B,\ o \not\leftrightarrow A_\xi) \le \tfrac{3}{4}\eta.$$

Step 4: the target. Condition on ξ. Then A_ξ is fixed, the remaining bits keep their product law, and $\mathbb{P}(a \leftrightarrow T \mid \xi) > 1 - \tau$ for every $a \in A_\xi$. Corollary 2.2, applied with the bits of ξ fixed, gives $\mathbb{P}(o \leftrightarrow A_\xi,\ o \not\leftrightarrow T \mid \xi) \le \tau < \eta/4$. Splitting according to whether $o \leftrightarrow A_\xi$ gives $\mathbb{P}(o \leftrightarrow B,\ o \not\leftrightarrow T) < \eta$. □

5 Relay between neighbouring coarse cubes

For $v = (v_1, v_2) \in \mathbb{Z}^2$ let

$$c_v = (20r+1)(v_1, v_2, 0, \dots, 0) \in \mathbb{Z}^d, \qquad Q_v = \Lambda_{10r}(c_v), \qquad B_v = \Lambda_{4r}(c_v).$$

The cubes Q_v are vertex-disjoint. For neighbours v, x in $\mathbb{Z}^2$, let I_{vx} be the finite set of pairs $\{y, y'\}$ with $y \in Q_v$, $y' \in Q_x$ and $J(y - y') > 0$. The union $D_{vx} = Q_v \cup Q_x$ is a rectangle with $|D_{vx}| = (40r+2)(20r+1)^{d-1} \le (41r+1)^d$, whose internal bits are those internal to Q_v, those internal to Q_x, and I_{vx}. Let D°_{vx} be the set of points of D_{vx} at distance more than $2r$ from $\mathbb{Z}^d \setminus D_{vx}$.

Lemma 5.1 (relay)**.** *Let v, x be neighbours in $\mathbb{Z}^2$ and G a finite graph as in Section 2 with $D = D_{vx} \subset V$ and $(F_q(D))$. Assume*

(E2°) every edge $\{y, z\}$ of G with $z \in D^\circ_{vx}$ and $y \notin D$ has $\lambda_{yz} \le qJ(y - z)$.

If $o \in V \setminus D$, then $\mathbb{P}(o \leftrightarrow B_v,\ o \not\leftrightarrow B_x) \le (d + 10)\eta$.

Proof. By (Sym) we may assume $c_v = 0$ and $c_x = (20r + 1)e_1$, so that $D = [-10r, 30r + 1] \times [-10r, 10r]^{d-1}$. Put $R' = R + w$. We define rectangles $P_0, \dots, P_{d+10}$ and apply Lemma 4.1 with $B = P_{k-1}$ and $T = P_k$; the radius is $2r$ for $k \le d + 9$ and $2r + 1$ for $k = d + 10$. Let $P_0 = B_v$. For $t = 1, \dots, d$ let P_t be centred at 0 with half-width $2r + tR'$ in directions $h \le t$ and $4r + tR'$ in directions $h > t$. For $i = 1, \dots, 10$ put $s_i = 2r$ $(i \le 9)$, $s_{10} = 2r + 1$, $z_i = (s_1 + \cdots + s_i)e_1$, $w_i = 2r + (d + i)R'$, and $P_{d+i} = \Lambda_{w_i}(z_i)$.

Contraction steps. Let $1 \le t \le d$ and $u \in P^{(R)}_{t-1}$. Then $|u_h| \le 2r + tR' - w$ for $h < t$ and $|u_h| \le 4r + tR' - w$ for $h \ge t$. Take $F = F_{t,\sigma,\varepsilon}(u, 2r)$ with $\sigma = -\operatorname{sign} u_t$ and $\varepsilon_j = -\operatorname{sign} u_j$ (with $\operatorname{sign} 0 = +1$). A point $y \in F$ has $y_t = u_t + \sigma s$ with $s \in (2r, 2r + w]$, so $|y_t| \le |u_t| - 2r \le 2r + tR'$ if $|u_t| \ge 2r + w$ and $|y_t| \le 2r + w$ otherwise. For $j \ne t$, y_j lies between u_j and $u_j - \operatorname{sign}(u_j)(2r + w)$, so $|y_j| \le \max\{|u_j|, 2r + w\}$, which is at most the half-width of P_t in direction j. Thus $F \subset P_t$. The coordinates of $\Lambda_{2r+w}(u)$ are at most $6r + dR' + w < 8r$ in absolute value, by (S6), so $\Lambda_{2r+w}(u) \subset Q_v$ and $P^{(R)}_{t-1} \subset D^\circ_{vx}$. After the last contraction, $P_d = \Lambda_{2r+dR'}(0)$.

Translation steps. Let $1 \le i \le 10$ and $u \in P^{(R)}_{d+i-1}$, so $\|u - z_{i-1}\| \le w_{i-1} + R$ with $w_0 = 2r + dR'$. Take $F = F_{1,+,\varepsilon}(u, s_i)$ with $\varepsilon_j = -\operatorname{sign} u_j$. For $y \in F$, $y_1 - (z_i)_1 = (u_1 - (z_{i-1})_1) + (y_1 - u_1 - s_i)$ with $y_1 - u_1 - s_i \in (0, w]$, so $|y_1 - (z_i)_1| \le w_{i-1} + R + w = w_i$, and $|y_j| \le \max\{|u_j|, s_i + w\} \le w_i$ for $j \ge 2$. Thus $F \subset P_{d+i}$. The first coordinates of $\Lambda_{s_i+w}(u)$ lie in $[-4r - (d + 11)R' - 1,\ 22r + (d + 11)R' + 1]$ and the others have absolute value at most $4r + (d + 11)R' + 1$; by (S6) these lie in D at distance more than $2r$ from its complement. Finally $z_{10} = c_x$ and $w_{10} = 2r + (d + 10)R' < 4r$, so $P_{d+10} \subset B_x$.

Each $P^{(R)}_{k-1}$ lies in D°_{vx}, so (E2°) gives (E2), and (E3) was just verified; also $o \notin D$. If $o \leftrightarrow P_0$ but $o \not\leftrightarrow P_{d+10}$, then $o \leftrightarrow P_{k-1}$ and $o \not\leftrightarrow P_k$ for some k, and the union bound over the $d + 10$ applications of Lemma 4.1 proves the claim. □

6 Exploration and the proof of Theorem 1.1

We follow the exploration of [17, §6], with the long-range interfaces I_{vx}. Let $o = 0 \in Q_0$, and let $\mathcal{I}$ be the event that all bits internal to Q_0 are open. It prescribes finitely many bits, each of probability in $(0, 1)$, so $\mathbb{P}_q(\mathcal{I}) > 0$, and under $\mathbb{P}_\mathcal{I} = \mathbb{P}_q(\cdot \mid \mathcal{I})$ all other bits are independent with their parameter-q law.

The exploration. The coarse root $0 \in \mathbb{Z}^2$ is declared good and visited, but is never queued or processed; its four neighbours are put in a queue with parent 0. A coarse vertex is *visited* once it has been queued, and *pending* until it is processed; vertices are processed in queue order, and none is queued twice. For a pending v, let $G(v)$ be the graph consisting of the cubes on the parent chain of v back to the root, the bits internal to these cubes, and the interfaces I between successive cubes of the chain, and no other pairs. Put $H_v = \{o \leftrightarrow B_v \text{ in } G(v)\}$. To process v, reveal the bits internal to Q_v and those of I_{pv}, where p is the parent of v. For each unvisited neighbour x of v, let $G' = G(v) \cup Q_x \cup I_{vx}$ (with the bits internal to Q_x) and compute

$$b_x = \mathbb{P}_\mathcal{I}(o \leftrightarrow B_x \text{ in } G' \mid \text{history after this reveal}),$$

which reveals nothing. Declare v *good* if H_v holds and $b_x > 1-\delta$ for every such x, and in that case queue these x with parent v; otherwise declare v *bad*. We maintain

$$\mathbb{P}_{\mathcal{I}}(H_v \mid \text{revealed history}) > 1-\delta \qquad \text{for every pending } v. \tag{6.1}$$

Lemma 6.1 (freshness and the initial promise)**.** *Conditional on any revealed history of positive probability, the bits that are neither prescribed by $\mathcal{I}$ nor revealed are independent with their parameter-q law, and (6.1) holds at every step.*

Proof. Each step reveals a finite set of bits determined by earlier revealed values, and all decisions are functions of revealed values; induction gives the first claim. The unknown bits of H_x for a pending x with parent v are internal to Q_x or lie in I_{vx}. The cubes are vertex-disjoint, each step reveals the internal bits of one cube and its own parent interface, and $G(x)$ contains no other pairs, so these bits stay unrevealed while x is pending; together with the definition of *good*, this keeps (6.1) for every queued x once it holds when x is queued.

It remains to check (6.1) for the root's neighbours. Let v be one; by symmetry Q_0 lies in direction $-e_1$ from Q_v. Let $\ell' = 10r - w \in \mathcal{L}$, $u = c_v - we_1$ and $F = F_{1,-,(+,\dots,+)}(u,\ell')$. Then $\Lambda_{\ell'}(u) \subset Q_v$; $\Lambda_m(u) \subset B_v$ since $m + w \le 4r$; and $F \subset Q_0$, since its first coordinates lie in $[(c_v)_1 - 10r - w, (c_v)_1 - 10r - 1]$ and its other coordinates in $[0, 10r]$. On $X(u,\ell',F)$, the box $\Lambda_m(u)$ is joined inside Q_v to a vertex with an open edge of length at most w into Q_0, and this edge lies in I_{0v}, so $X(u,\ell',F) \cap \mathcal{I} \subset H_v$ (the cube Q_0 is open and connected by (Pos)). The event $X(u,\ell',F)$ depends only on bits internal to Q_v and on pairs of I_{0v}, which are independent of $\mathcal{I}$; hence $\mathbb{P}_{\mathcal{I}}(H_v) \ge \mathbb{P}_q(X(u,\ell',F)) > 1-\gamma > 1-\delta$ by (S7) and (S2). □

Lemma 6.2. *Immediately before a pending v is processed, the conditional probability that v is declared bad is less than f.*

Proof. Fix an unvisited neighbour x of v and let $D = D_{vx}$. Neither v (pending) nor x (unvisited) has been processed and I_{vx} is not a parent interface yet, so every bit internal to D is fresh with parameter q, and $(F_q(D))$ holds in G'. The pairs of G' with one endpoint in D and the other outside are those of I_{pv}, which is still unrevealed, with $\lambda = qJ$; so (E2°) holds. Since $o \notin D$, Lemma 5.1 gives $\mathbb{P}_{\mathcal{I}}(H_v \cap \{o \not\leftrightarrow B_x \text{ in } G'\} \mid \text{history}) \le (d+10)\eta$, and the tower property gives $\mathbb{P}_{\mathcal{I}}(H_v,\ b_x \le 1-\delta \mid \text{history}) \le (d+10)\eta/\delta$. With at most four such x and (6.1), $\mathbb{P}_{\mathcal{I}}(v \text{ bad} \mid \text{history}) \le \delta + 4(d+10)\eta/\delta < f$ by (S2). □

Lemma 6.3. *For every deterministic set of k distinct coarse vertices, the $\mathbb{P}_{\mathcal{I}}$-probability that all of them are processed and declared bad is at most f^k.*

Proof. The root is good, so assume it is not in the set S. Let τ_j be the time at which the jth member of S is processed ($\tau_j = \infty$ if there is none) and E_j the event that $\tau_j < \infty$ and the first j processed members are bad. For each finite t the event $\{\tau_j = t\} \cap E_{j-1}$ is determined before the reveal at time t, so Lemma 6.2 gives $\mathbb{P}_{\mathcal{I}}(E_j) \le f \sum_t \mathbb{P}_{\mathcal{I}}(\tau_j = t, E_{j-1}) \le f\,\mathbb{P}_{\mathcal{I}}(E_{j-1})$, and induction finishes the proof. □

Proof of Theorem 1.1. By Lemma 2.4, $0 < \beta_c < \infty$. Suppose that $\theta(\beta_c) > 0$ and choose the parameters of Lemma 3.3 with $\beta = \beta_c$. Let $X \subset \mathbb{Z}^2$ be the set of good coarse vertices. If X is finite, the queue empties after finitely many steps, and every coarse vertex outside X adjacent to X has been processed and declared bad. Apply Lemma 2.3 to X: its boundary contains a cycle of some length n from a family of at most $2n3^n$ candidates, each with $k \ge n/7$ crossed edges with disjoint endpoints, each of which has a bad endpoint outside X. Choosing one endpoint per edge

in 2^k ways, Lemma 6.3 bounds the probability of a given candidate by $(2f)^k \le (2f)^{n/7}$. By (S1), $\mathbb{P}_{\mathcal{I}}(|X| < \infty) \le \sum_n 2n3^n(2f)^{n/7} < 1$. Every good non-root vertex v satisfies H_v, and the boxes B_v are disjoint, so with positive $\mathbb{P}_{\mathcal{I}}$-probability the open cluster of o in the long-range model is infinite. Hence $\theta(q) \ge \mathbb{P}_q(\mathcal{I})\mathbb{P}_{\mathcal{I}}(|X| = \infty) > 0$ with $q < \beta_c$, a contradiction. □

7 The corollaries, and where the hypotheses enter

Proof of Corollary 1.2. If $\alpha > d$ then $T(n) \le C'n^{-\alpha}$, so (Tail) holds, and (Pos) holds by the lower bound; apply Theorem 1.1. If $\alpha < d$, put $s = d + \alpha < 2d$ and $P_k = 1 - e^{-\beta J(k)}$ at a parameter β with $\theta(\beta) > 0$. Berger's theorem [3, Theorem 2.6] states: if $\{P_k\}$ is percolating and $\liminf_{\|k\|\to\infty} P_k\|k\|_1^s > 0$ for some $s < 2d$, then $\{(1-\epsilon)P_k\}$ is percolating for some $\epsilon > 0$ (his standing assumption is $0 \le P_k = P_{-k} < 1$, which holds here). Monotonicity and $1 - e^{-t} \ge t/2$ on $[0, 1]$ give $P_k \ge (\beta c/2)\|k\|^{-s} \ge (\beta c/2)\|k\|_1^{-s}$ for large $\|k\|$. Concavity of $t \mapsto 1 - e^{-t}$ gives $1 - e^{-(1-\epsilon)\beta J} \ge (1-\epsilon)(1 - e^{-\beta J})$, so $\theta((1-\epsilon)\beta) > 0$. Thus $\theta(\beta) > 0$ forces $\beta > \beta_c$. □

Proof of Corollary 1.3. Exponential decay and finite range both give $n^d T(n) \to 0$. □

Remark 7.1 (where the hypotheses enter). The tail condition is used only in (S5), at one pair of scales (w, r) with r a large constant multiple of w fixed by (S6): in Lemma 3.2, for exits from Λ_ℓ by edges longer than w, and in the event E_{long} of Lemma 4.1, for entrances that jump over the shell. The layer argument needs $R \gg e^L w$ and the relay needs $r \gg R + w$; both hold with $r \le C_0 w$, and (Tail) then makes $\beta(41r + 1)^d T(w) \le \beta(42C_0)^d w^d T(w)$ small. Summability gives the uniform bound $a_z \le A$, which turns a large crossing weight into many seed attempts. (Sym) gives the equal probabilities of the faces in Lemma 3.2 and the orientation reductions. (Pos) gives $\beta_c < \infty$ and the connectivity of open boxes. The dimension $d \ge 2$ enters through the coordinate plane in Lemma 2.4 and the coarse lattice $\mathbb{Z}^2$ with its contour count.

Remark 7.2 (the borderline). For $J(x) \asymp \|x\|^{-2d}$ the quantity $n^d T(n)$ stays bounded away from zero, and nothing in the argument guarantees (S5) and (S6) together: long edges at scale n leave the cube $[-n, n]^d$ with probability of order one at every scale. In $d = 1$ and $J(x) \asymp |x|^{-2}$ the transition is discontinuous [1, 5], and the argument above also lacks the coarse plane. Apart from sufficiently spread-out kernels in $d \ge 7$, where $d > 6 = 3(\alpha \wedge 2)$ and [12] applies, whether $\theta(\beta_c) = 0$ for $\alpha = d \ge 2$ remains open.

8 Closing discussion and open problems

The borderline exponent. Remark 7.2 explains why the argument stops at $\alpha = d$: long edges leave cubes with probability of order one at every scale, so neither the long-edge term of Lemma 3.2 nor the long entrances of Lemma 4.1 can be made small.

Question 8.1. Let $d \ge 2$ and let J satisfy (Sym) and $c\|x\|^{-2d} \le J(x) \le C\|x\|^{-2d}$ for all $x \ne 0$, with $0 < c \le C < \infty$. Is $\theta(\beta_c) = 0$?

In $d = 1$ the transition for $J(x) \asymp |x|^{-2}$ is discontinuous [1, 5], so a positive answer has to use $d \ge 2$.

Symmetry. By Remark 7.1, (Sym) is used only to give the faces of Lemma 3.2 equal probabilities and to reduce the relay to one orientation, and (Pos) only for $\beta_c < \infty$ and for the connectivity of open boxes.

Question 8.2. Let $d \geq 2$ and let $J(x) = J(-x)$ satisfy (Tail), with support generating $\mathbb{Z}^d$. Is $\theta(\beta_c) = 0$?

Quantitative bounds. The proof works by contradiction: from $\theta(\beta_c) > 0$ it produces $q < \beta_c$ with $\theta(q) > 0$. It therefore gives no bound on $\theta(\beta)$ for $\beta > \beta_c$, nor on $\mathbb{P}_{\beta_c}(0 \leftrightarrow \mathbb{Z}^d \setminus \Lambda_n)$, and the renormalization argument of [17] that it adapts is qualitative in the same way.

Question 8.3. Under the hypotheses of Theorem 1.1, are there $c, C > 0$ such that $\theta(\beta) \leq C(\beta - \beta_c)^c$ for all $\beta > \beta_c$ close to β_c?

Formalization. The Lean proof described in Section 1.4 covers Theorem 1.1. Corollary 1.3 and the case $\alpha > d$ of Corollary 1.2 follow from it by bounding $T(n)$. The case $\alpha < d$ would need a formalization of Berger's theorem [3].

References

[1] M. Aizenman and C. M. Newman. Discontinuity of the percolation density in one-dimensional $1/|x-y|^2$ percolation models. *Comm. Math. Phys.*, 107(4):611–647, 1986.

[2] O. Angel, D. de la Riva, C. Panagiotis, and Y. Shi. Continuity of the phase transition for long-range Bernoulli percolation with exponential decay on $\mathbb{Z}^2$. Preprint, arXiv:2610.35233, 2026.

[3] N. Berger. Transience, recurrence and critical behavior for long-range percolation. *Comm. Math. Phys.*, 226:531–558, 2002. doi:10.1007/s002200200617, arXiv:math/0110296.

[4] A. Blanc-Renaudie. Notes on Kozma and Nitzan's proof of $\theta(p_c) = 0$. Preprint, arXiv:2610.10502, 2026.

[5] H. Duminil-Copin, C. Garban, and V. Tassion. Long-range models in 1D revisited. Preprint, arXiv:2011.04642, 2020.

[6] H. Duminil-Copin, V. Sidoravicius, and V. Tassion. Absence of infinite cluster for critical Bernoulli percolation on slabs. *Comm. Pure Appl. Math.*, 69(7):1397–1411, 2016.

[7] R. Fitzner and R. van der Hofstad. Mean-field behavior for nearest-neighbor percolation in $d > 10$. *Electron. J. Probab.*, 22:43, 2017. doi:10.1214/17-EJP56.

[8] J.-B. Gouéré. On the proof of the Kozma–Nitzan conjecture. Preprint, arXiv:2610.30063, 2026.

[9] G. R. Grimmett and J. M. Marstrand. The supercritical phase of percolation is well behaved. *Proc. Roy. Soc. London Ser. A*, 430(1879):439–457, 1990.

[10] T. Hara and G. Slade. Mean-field critical behaviour for percolation in high dimensions. *Comm. Math. Phys.*, 128(2):333–391, 1990.

[11] T. E. Harris. A lower bound for the critical probability in a certain percolation process. *Proc. Cambridge Philos. Soc.*, 56:13–20, 1960.

[12] M. Heydenreich, R. van der Hofstad, and A. Sakai. Mean-field behavior for long- and finite range Ising model, percolation and self-avoiding walk. *J. Stat. Phys.*, 132:1001–1049, 2008. doi:10.1007/s10955-008-9580-5.

[13] T. Hutchcroft. Power-law bounds for critical long-range percolation below the upper-critical dimension. *Probab. Theory Related Fields*, 181:533–570, 2021. arXiv:2008.11197.

[14] H. Kesten. The critical probability of bond percolation on the square lattice equals 1/2. *Comm. Math. Phys.*, 74:41–59, 1980.

[15] G. Kozma and S. Nitzan. A reduction of the $\theta(p_c) = 0$ problem to a conjectured inequality. Preprint, arXiv:2401.12397, 2024.

[16] J. Leder. $\theta(p_c) = 0$ for Bernoulli bond percolation on $\mathbb{Z}^d$ in all dimensions $d \geq 2$, in Lean 4/Mathlib, 2026. https://github.com/anthropics/formal-math/tree/795efb86f191735c5481675763537cfb4ff37e55/percolation.

[17] OpenAI. Critical bond and site percolation on the cubic lattice, 2026. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Critical-bond-and-site-percolation-on-the-cubic-lattice-September-24-2026.

[18] OpenAI. No percolation at criticality on quasi-transitive graphs, 2026. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/No-percolation-at-criticality-on-quasi-transitive-graphs-September-24-2026.

[19] Y. R. Pei. Lean 4 formalization of Theorem 1.1, 2026. https://github.com/PeaBrane/lr-continuity-lean.